

TAHA ELZAKI

GIS ANALYST & Remote sensing Specialist

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PROFESSIONAL SUMMARY

GIS Analyst with extensive experience in **geospatial data** extraction, **remote sensing**, object-based image analysis (**OBIA**), morphometrics, and **machine learning/deep learning** for feature and building classification. Skilled in integrating multispectral imagery, and field survey data to produce high-accuracy spatial products for **urban planning**, infrastructure mapping, and environmental assessment. **QGIS** certified, with strong command of **ArcGIS Pro**, **ENVI**, **PostGIS**, and advanced geospatial modelling workflows.

LANGUAGES AND AVAILABILITY

- **Arabic:** Native · **English:** Advanced
- **Availability:** Immediate for employment in Saudi Arabia

SKILLS

- **Geospatial Platforms:** ArcGIS Pro; ERDAS Imagine; QGIS; PostGIS; Global Mapper.
- **Remote Sensing and OBIA:** ENVI; ERDAS; Google Earth Engine; object-based image analysis for segmentation and classification; spectral indices.
- **Spatial Analysis:** deprivation analysis, indices, raster/vector processing
- **Programming and Data Management:** Python; SQL; JavaScript (GEE); spatial databases.
- **Field Survey :** GNSS/GPS; total station
- **Soft Skills:** Communication, teamwork, adaptability.

EXPERIENCE

Federal Ministry of Transport - GIS Specialist, Khartoum, Sudan, Jan 2020 – Jun 2024

- Developed and maintained GIS databases to support transportation planning initiatives.
- Analyzed spatial data using ArcGIS to inform decision-making processes for federal transport projects.
- Collaborated with multidisciplinary teams to integrate GIS technologies into project workflows and operational strategies.
- Conducted spatial analysis for roads, drainage, and flood-risk areas.

WORLD FOOD PROGRAM(WFP) - VOLUNTEER PERSON SUPPORT,

North Darfur, Sudan, Sept 2022 - Mar 2023

- provided comprehensive support for program implementation, enhancing operational efficiency and collaboration across teams.
- Assisted in data collection and analysis, contributing to informed decision-making and strategic planning initiatives.

UNIVERSITY OF TWENTE - GIS Specialist, Enschede, Netherlands, Mar 2021 - Mar 2023

- Implemented a Deprivation Area analysis system combining remote sensing classification, socioeconomic indicators, and displacement mapping.
- Applied object-based image analysis and deep learning segmentation to increase building and settlement classification reliability.

River Navigation Authority - Survey Engineer, Khartoum, Sudan, Feb 2018 - Dec 2019

- Performed land and hydrographic surveys; integrated GNSS and total station outputs into GIS for navigation planning and boundary delineation.
- Developed and maintained a comprehensive database of survey data, facilitating easy access and analysis for future projects.

AGRESA – GIS Survey, Khartoum, Sudan, JULY 2015 - Dec 2015

- Conducted field surveys and processed spatial data for agricultural mapping and land assessment projects.

PROJECTS & PUBLICATIONS

- Collaborated with team of IdeamapSudan: Geo-spatial for urban poverty Urban Remote Sensing Event (JURSE), <https://ieeexplore.ieee.org/document/10144211>.
- <https://research.utwente.nl/en/publications/ideamapsudan-geo-spatial-modelling-of-urban-poverty>.

EDUCATION & CERTIFICATIONS

- **MSc Geoinformatics** - Future University, Khartoum, 2020
- **BSc Survey Engineering** - Sudan University of Science & Technology, 2015
- **QGIS Certification** - Data management; Raster & vector analysis; Image classification, <https://certification.qgis.org/en/certificate/QGIS-3113/>
- **IdeamapSudan** Integrated Deprivation Area mapping system for Displacement Durable Solutions and socioeconomic reconstruction in Khartoum, University of Twente Faculty of Geo-Information Science and Earth Observation.2023
<https://www.idea-maps.net/>
- **Courses:** Python (2024); Jupyter; Data Science; Agile Project Management & Data analysis.

REFERENCES

Available on request.